

Short Answer Test

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Sub Name: Computer Vision

① Computer Vision:- It uses Image Processing, Feature extraction and machine Learning algorithms to analyze digital images and videos, identify objects, recognize patterns and by following this method it understands visual information.

a) main objective of computer vision:- Image understanding, preprocess sensing object, converting the data, changing the image and decision making based on image.

b) Real world Applications:-

⇒ Autonomous vehicles

⇒ medical image diagnosis

② The main features are image acquisition, Feature scanning, object sensing, making decisions based on their image size and shape and colour; It will also do image segmentation.

a) Feature Extraction:- Feature Extraction means from the given image identifying important characteristics like edges, corners, textures, shapes from an image for analysis.

b) How does the object recognition improve computer vision system.

A) If it is improved then making the decision based on image or video will be computer vision system will give a perfect decision.

③ There are three types of levels one is Low-Level, Mid-Level, High Level.

④ Low-Level image Processing:- Deals with basic operations such as noise removal, image enhancement, filtering.

⑥ Difference b/w mid and high levels?

A) Mid Level Image Processing:-

i) Segmentation and Feature extraction

ii) Produces image features

High Level Processing

i) Object recognition and scene understanding

ii) Produces meaningful decisions

④ Digital image is Represented in matrix of Pixels Each Pixels store intensity or colour of matrix information

a) Pixel:- It is a smallest element in Digital image which stores colour and brightness of image.

b) how does the image resolution affect image quality?

A) higher ~~Pixels~~ resolution means more no. of Pixels and the image will be more clarity with better detail.

⑤ An image sensor captures light from a real-world scene and converts it into electrical signal, which are digitized into digital image.

a) image sensor:- image sensor captures the real-time image and converts light energy to electrical signal

⑥ Types of image acquisition devices?

i) Digital camera

ii) CCD/CMOS camera

⑦ Sampling converts a continuous image into discrete image. while quantization distributes the intensity values for those Pixels.

a) Sampling:- Sampling is the Process of dividing a continuous image into discrete image Pixels.

⑥ Purpose of Quantization:

A) To assign finite digital intensity values to sampled pixels for storage and further processing.

⑦ It explains how light from scene passes through a camera lens to form an image on the sensor

a) Perspective Projection:

A) It is the projection of an 3D scene into a 2D image plane, making distant image appear smaller.

b) Role of Illumination:

A) illumination provides the light necessary to capture clear and accurate images

⑧ a) Define image processing: image processing is the technique of enhancing or modifying digital images

⑥ Two differences:

Image Processing	Computer vision
Enhances images	understands image content
output is an improved image	output is information or decisions

⇒ how are they related means.

Computer vision: Image processing it enhances or transforms that image.

Now computer vision takes that processed image and it interprets and understands image.

⑨ Computer vision is widely used in healthcare, autonomous ~~vehicles~~ vehicles, surveillance, robotics, and industrial automation.

⑤ Healthcare application:- Detects diseases from x-rays, CT-scans, and MRI images for early diagnosis.

⑥ Autonomous vehicle application:- Detects lanes, traffic signs, pedestrians.

⑩ Digital image fundamentals provide accurate image representation, enabling reliable analysis feature extraction, and object recognition.

④ Why are Pixels the basic building blocks?

A) Because every digital image is made up of Pixels, each storing brightness or colour information

⑤ Sampling and resolution influence?

A) Higher sampling and resolution capture more details improving feature extraction and recognition accuracy